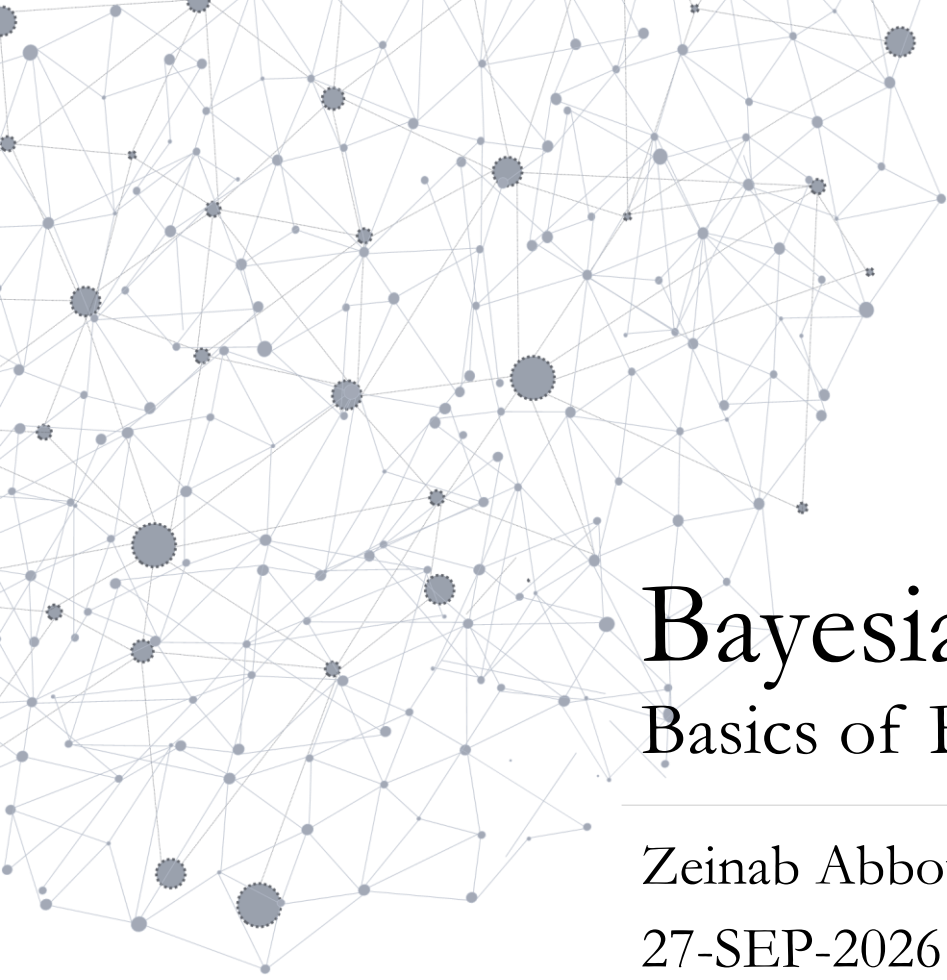




Bayesian Neural Networks

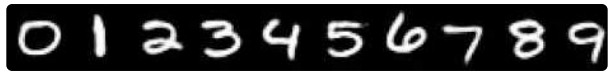
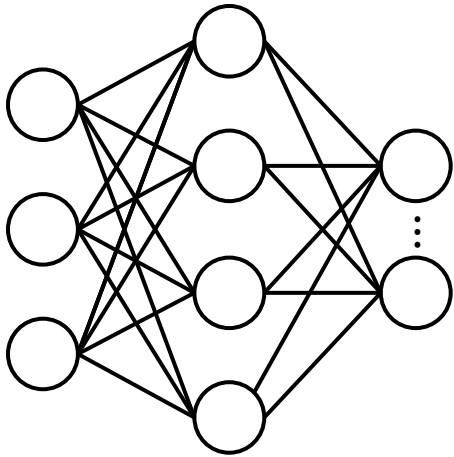
Basics of Bayesian Models of Uncertainty

Zeinab Abboud

27-SEP-2026

Problem with deterministic neural networks

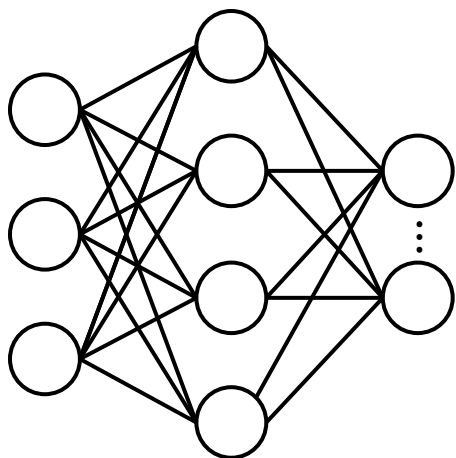
MLP Trained on MNIST



Test Accuracy $\cong 99\%$

Problem with deterministic neural networks

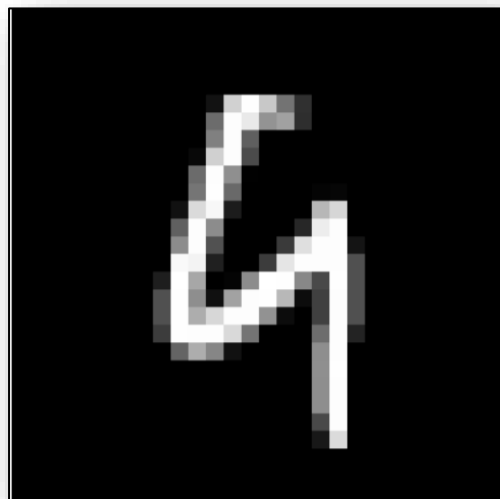
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

Test Accuracy $\cong 99\%$

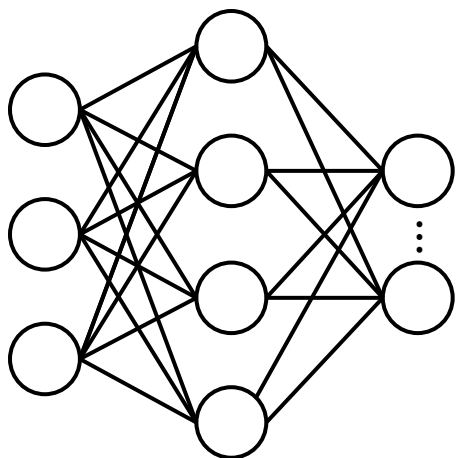
Ambiguous input image



Is it a 4 ? a 9?

Problem with deterministic neural networks

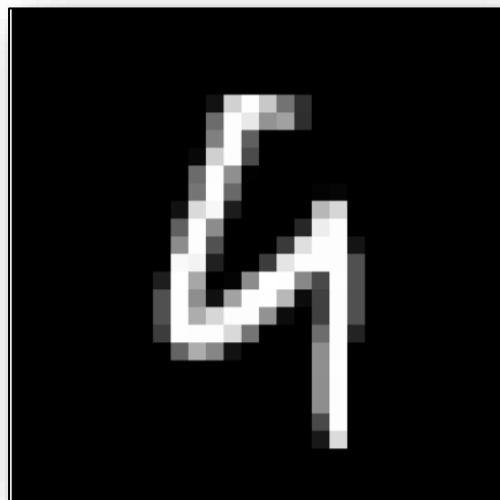
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

Test Accuracy $\cong 99\%$

Ambiguous input image

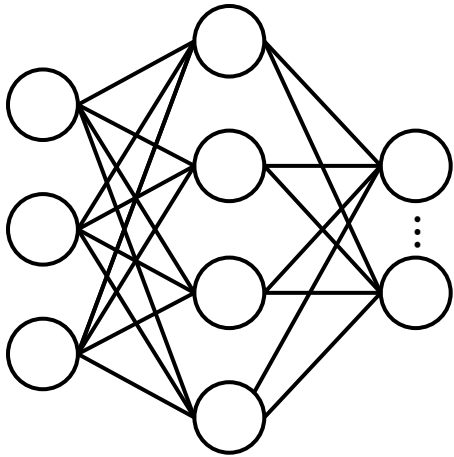


Is it a 4 ? a 9?

MLP model predicts
 $p(4|\text{image}) = 0.99$

Problem with deterministic neural networks

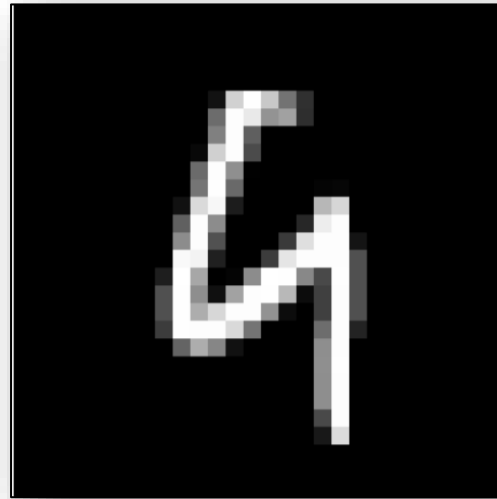
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

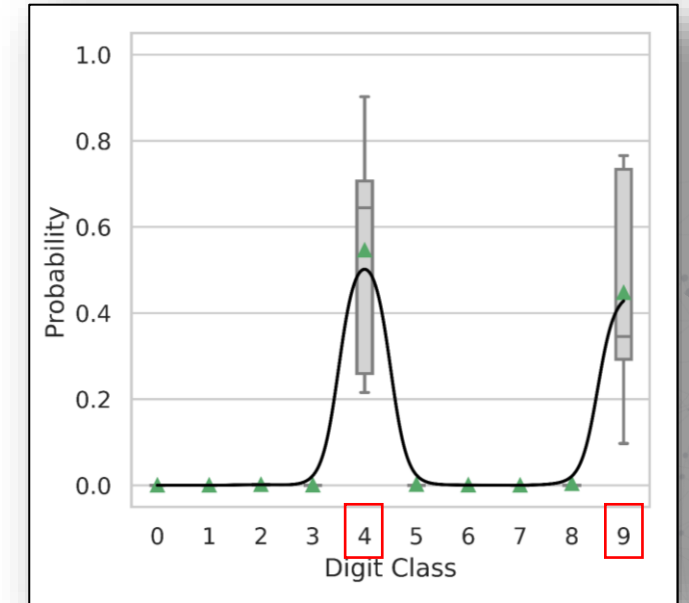
Test Accuracy $\cong 99\%$

Ambiguous input image



Is it a 4 ? a 9?

MLP model predicts
 $p(4|\text{4}) = 0.99$



A more realistic
output of $p(y|x)$

Outline

- Recap of deterministic neural network
- Bayes' theorem
- Application of Bayes' theorem in NN
- Hands on demo

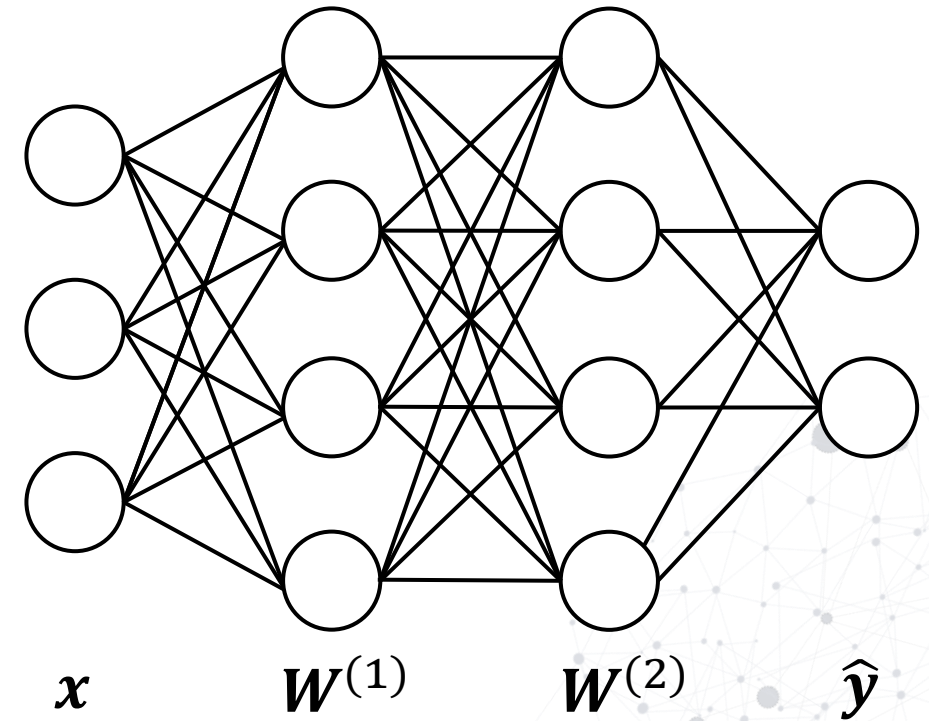
Neural Networksback to basics...

Data $\rightarrow D = \mathbf{x}^n, \mathbf{y}^n$

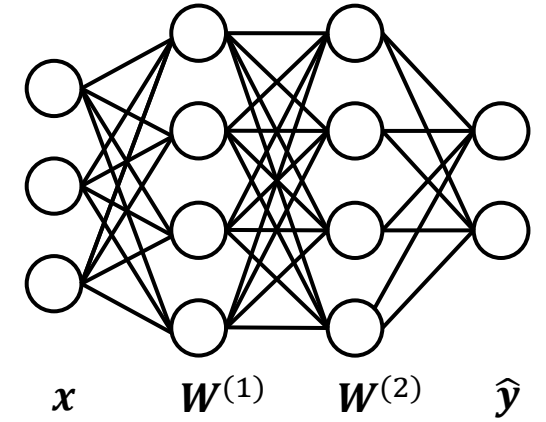
Parameters $\rightarrow \theta = \{w, b\}$

Model $\rightarrow p(\mathbf{y}|\mathbf{x}, \theta) \leftarrow$ likelihood

Output $\rightarrow \hat{\mathbf{y}} = W^{(2)}\sigma(W^{(1)}\mathbf{x} + b)$



Neural Networksback to basics...



Data $\rightarrow D = x^n, y^n$

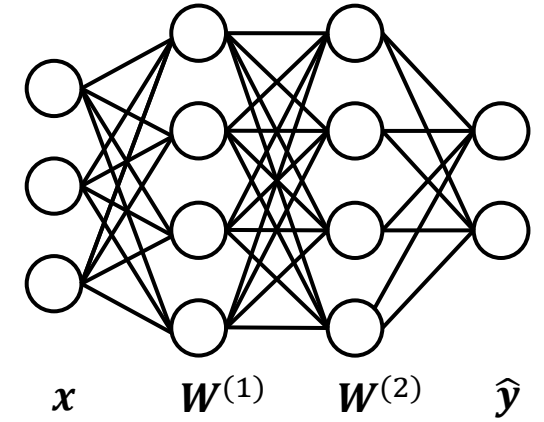
Parameters $\rightarrow \theta = \{w, b\}$

Model $\rightarrow p(y|x, \theta)$

Output $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

$$\hat{y} = \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \dots & w_{1H}^{(2)} \\ w_{H1}^{(2)} & \dots & w_{HN}^{(2)} \end{bmatrix}}_{W^{(2)}} \sigma \left(\underbrace{\begin{bmatrix} w_{11}^{(1)} & \dots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \dots & w_{HN}^{(1)} \end{bmatrix}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}}_{b} \right)$$

Neural Networksback to basics...



Data $\rightarrow D = x^n, y^n$

Parameters $\rightarrow \theta = \{w, b\}$

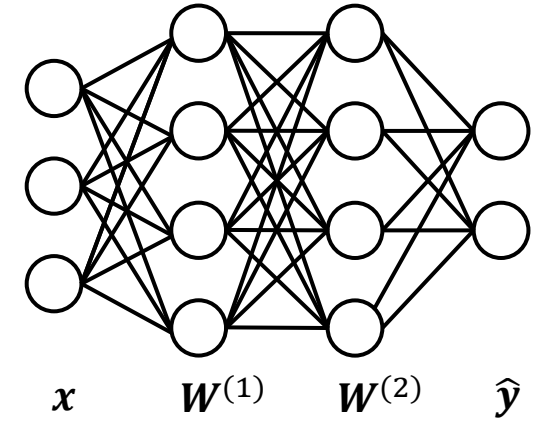
Model $\rightarrow p(y|x, \theta)$

Output $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

Point estimates

$$\hat{y} = \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \dots & w_{1H}^{(2)} \\ \vdots & & \vdots \\ w_{H1}^{(2)} & \dots & w_{HN}^{(2)} \end{bmatrix}}_{W^{(2)}} \sigma \left(\underbrace{\begin{bmatrix} w_{11}^{(1)} & \dots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \dots & w_{HN}^{(1)} \end{bmatrix}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}}_{b} \right)$$

Neural Networksback to basics...



Data $\rightarrow D = x^n, y^n$

Parameters $\rightarrow \theta = \{w, b\}$

Model $\rightarrow p(y|x, \theta)$

Output $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

Point estimates

$$\hat{y} = \underbrace{\begin{bmatrix} 0.01 & \dots & 0.04 \end{bmatrix}}_{W^{(2)}}_{(1,H)} \sigma \left(\underbrace{\begin{bmatrix} 0.01 & \dots & 0.03 \\ \vdots & \ddots & \vdots \\ 0.02 & \dots & 0.02 \end{bmatrix}}_{W^{(1)}}_{(H,N)} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}}_x_{(N,1)} + \underbrace{\begin{bmatrix} 0.1 \\ \vdots \\ 0.2 \end{bmatrix}}_b_{(H,1)} \right)$$

Fixed output y
given input x

Let's re-imagine the weights as distributions

$$\begin{aligned}
 & \begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)} \\
 & \quad \uparrow \\
 \hat{y} &= \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \cdots & w_{1H}^{(2)} \end{bmatrix}_{(1,H)}}_{W^{(2)}} \sigma \left(\underbrace{\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}_{(N,1)}}_x + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}_{(H,1)}}_b \right)
 \end{aligned}$$

Let's re-imagine the weights as distributions

Diagram illustrating the learning of parameters for a Gaussian distribution. The main matrix represents the distribution parameters:

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)}$$

Red arrows indicate the learning of parameters from the data matrix $W^{(1)}$ and the input vector x and bias vector b .

The parameters are learned from the data matrix $W^{(1)}$ and the input vector x and bias vector b :

$$\begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$

$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Learn the parameters of the distribution

The diagram also shows the relationship between the parameters and the data matrix $W^{(1)}$, the input vector x , and the bias vector b :

$$\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$

$$\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}_{(N,1)}$$

$$\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}_{(H,1)}$$

The diagram illustrates the learning of parameters for a Gaussian distribution. The main matrix represents the distribution parameters, which are learned from the data matrix $W^{(1)}$ and the input vector x and bias vector b . The parameters are the mean and variance, which are learned from the data. The diagram also shows the relationship between the parameters and the data matrix $W^{(1)}$, the input vector x , and the bias vector b .

Let's re-imagine the weights as distributions

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)}$$

Learn the parameters of the distribution

$$\begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$

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Now what?

How do we learn these parameters?

Bayesian Neural Networks

- Rethink of the neural network as a method to solve Bayes' theorem

$$p(\theta|x, y) = \frac{p(y|x, \theta) p(\theta)}{p(D)}$$

- Learn a posterior distribution $p(\theta|x, y)$ given a prior $p(\theta)$ over the neural network parameters.

Bayesian Neural Networks

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- Learn a posterior distribution $p(\theta|x, y)$ given a prior $p(\theta)$ over the neural network parameters.

What does learning a posterior mean?

Bayes' Theorem

$$p(A|B) = \frac{p(B|A)P(A)}{p(B)}$$



Bayes' Theorem

$$p(A|B) = \frac{p(B|A)P(A)}{p(B)}$$

Let's break this down...

Axioms of probability:

- 1) $0 \leq p(A) \leq 1$ with $p(A) = 1$ if A is certain
- 2) If events A_i are pairwise incompatible, then $p(\bigcup_i A_i) = \sum_i p(A_i)$
- 3) $p(A \cap B) = p(B|A)p(A)$

Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

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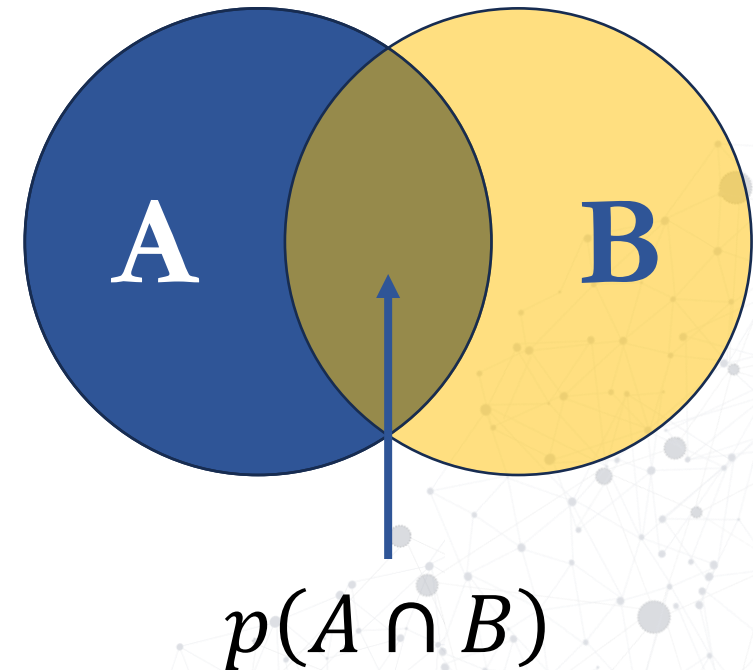
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Let's break this down...

Axioms of probability:

$$3) p(A \cap B) = p(B|A)p(A)$$



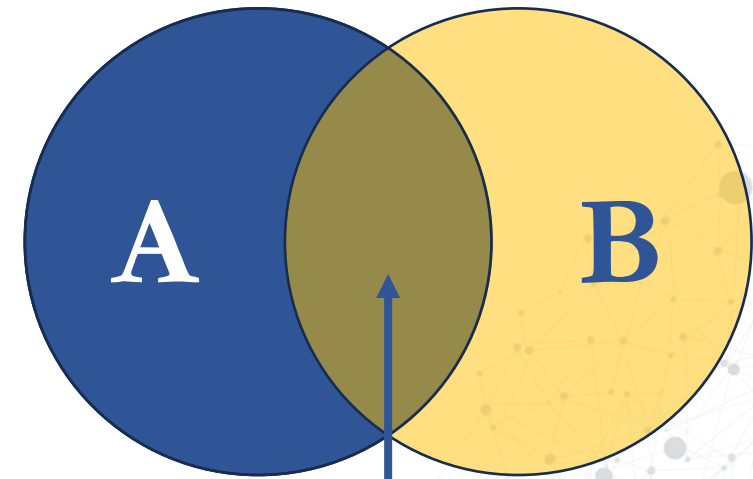
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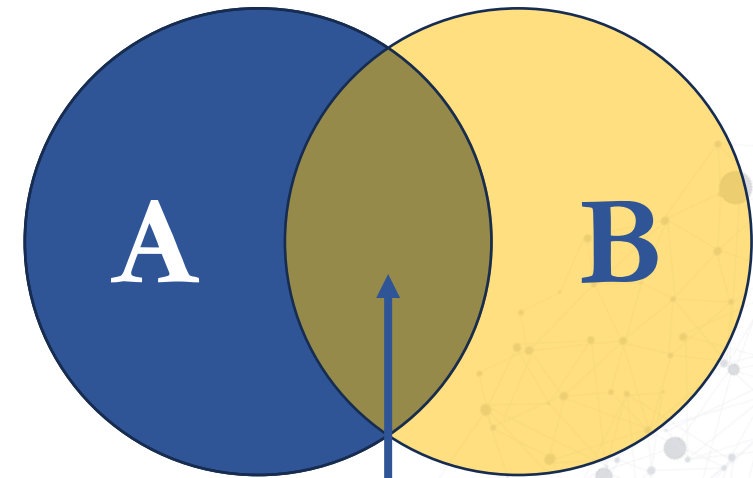
$p(A \cap B)$ or $p(B \cap A)$
or $p(A, B)$

Bayes' Theorem

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Let's break this down...

$$p(A \cap B) = p(B|A)p(A)$$
$$p(B \cap A) = p(A|B)p(B)$$



$p(A \cap B)$ or $p(B \cap A)$
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Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

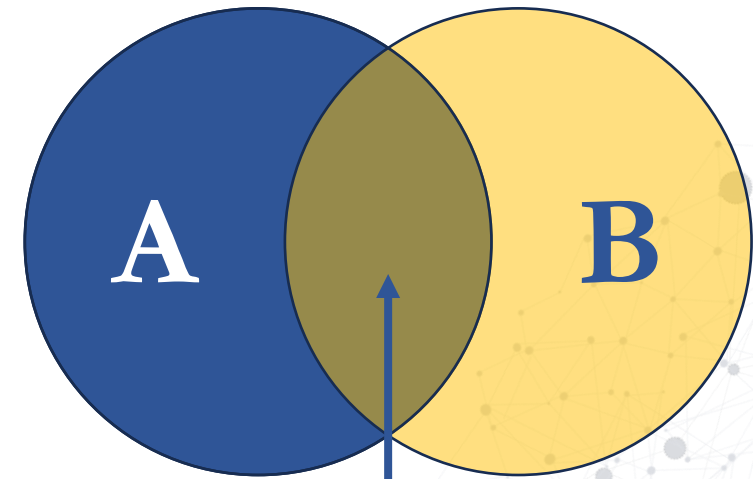
Let's break this down...

$$p(A \cap B) = p(B|A)p(A) \dots (1)$$

$$p(B \cap A) = p(A|B)p(B) \dots (2)$$

$$\because (1) = (2)$$

$$\therefore p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$



$p(A \cap B)$ or $p(B \cap A)$
or $p(A, B)$

Bayes' Theorem

$$p(A|B) = \frac{p(B|A)P(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease 

Precision of the test is 99%

Prevalence of  in population is 0.1%

Q: what is the probability that you have disease 
given that you tested positive?

Bayes' Theorem

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Bayes' Theorem

$$p(B) = p(B|A)p(A) + p(B|\sim A)p(\sim A)$$

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

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Prevalence of  in population is 0.1% $p(A)$

Q: what is the probability that you have disease 
given that you tested positive? $p(A|B)$

Bayes' Theorem

$$p(B) = p(B|A)p(A) + p(B|\sim A)p(\sim A)$$

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = 0.09$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

Precision of the test is 99% $p(B|A)$

Prevalence of  in population is 0.1% $p(A)$

Q: what is the probability that you have disease 
given that you tested positive? $p(A|B)$

Bayes' Theorem

$$p(A|B) = \frac{p(B|A)P(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease $p(B)$

Precision of the test is 99% $p(B|A)$

Prevalence of the disease in population is 0.1% $p(A)$

Q: what is the probability that you have disease
given that you tested positive? $p(A|B)$

Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive

Bayes' Theorem

$$p_2(A|B) = \frac{p(B|A)p_1(A|B)}{p_1(B)} = 0.91$$

Let's say you took a medical test, and tested positive for disease $p(B)$

Precision of the test is 99% $p(B|A)$

Prevalence of the disease in population is 0.1% $p(A) = p_1(A|B) = 0.09$

Q: what is the probability that you have disease
given that you tested positive? $p(A|B)$

Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

Bayes' Theorem

$$p_3(A|B) = \frac{p(B|A)p_2(A|B)}{p_2(B)} = 0.999$$

Let's say you took a medical test, and tested positive for disease $p(B)$

Precision of the test is 99% $p(B|A)$

Prevalence of the disease in population is 0.1% $p(A) = p_2(A|B) = 0.91$

Q: what is the probability that you have disease
given that you tested positive? $p(A|B)$

Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

Let's say you're still doubtful that you have the disease, you take another test, and it's positive...

Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p_{i-1}(A|B)}{p(B)} = 0.91$$

Let's say you took a medical test, and tested positive for disease $p(B)$

Bayes' theorem allows us to update the conditional probability of an event occurring when new data is presented

Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

How does Bayes' Theorem allow us to estimate uncertainty?

Bayes' Theorem

$$p(A|B) = \frac{p(B|A) P(A)}{p(B)}$$

Posterior

Likelihood Prior

Evidence
(Marginal)

Bayes' Rule for Neural Networks

$$p(\theta|x, y) = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{P(\theta)}}{\underset{\substack{\text{Evidence} \\ \text{(Marginal)}}}{p(D)}}$$

Posterior

Bayes' Theorem

$$p(A|B) = \frac{\overset{\text{Likelihood}}{p(B|A)} \overset{\text{Prior}}{P(A)}}{\underset{\substack{\text{Evidence} \\ \text{(Marginal)}}}{p(B)}}$$

Posterior

Bayes' Rule for Neural Networks

$$p(\theta|x, y) = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{P(\theta)}}{\underset{\text{Evidence (Marginal)}}{p(D)}}$$

Posterior

Evidence
(Marginal)

$$p(D) = \int p(D|\theta)p(\theta)d\theta$$

In high dimensional parameter space θ this integral is intractable (i.e. incomputable)

$$p(D) = \int_{\theta_0} \cdots \int_{\theta_{D-1}} p(X, \theta) d\theta_0 \cdots d\theta_{D-1}$$

Variational Inference for BNNs

approximating a posterior

$$\underset{\text{Posterior}}{p(\theta|x, y)} = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{P(\theta)}}{\underset{\substack{\text{Evidence} \\ \text{(Marginal)}}}{p(D)}} \longrightarrow \underset{\substack{\text{Approximate} \\ \text{Posterior}}}{q_{\phi}(\theta)} \approx p(\theta|x, y)$$

Variational Inference for BNNs

approximating a posterior

- Introduce a tractable family of distributions $q_{\phi}(\theta) = \prod q(\theta_i)$ (*)
Diagonal Gaussian with mean μ and variance σ^2
- Now the optimization criterion becomes of the following form

$$\mathcal{L}_{ELBO} = - \mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_{\phi}(\theta) || p(\theta))$$

Likelihood
Data-dependent

KL divergence
Prior-dependent

*with mean-field assumption, which assumes that all the parameters within the network are independent random variables

Let's re-imagine the weights as distributions

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)}$$

$\rightarrow$

$$\begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$
Learn the parameters of the distribution

$\rightarrow$

$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Now what?

How do we learn these parameters?

$\hat{y}$

$W^{(2)}$

$W^{(1)}$

x

b

$(H,1)$

How do we learn these distributions?

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)} \begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)} \begin{matrix} \text{Learn the} \\ \text{parameters of} \\ \text{the distribution} \end{matrix}$$

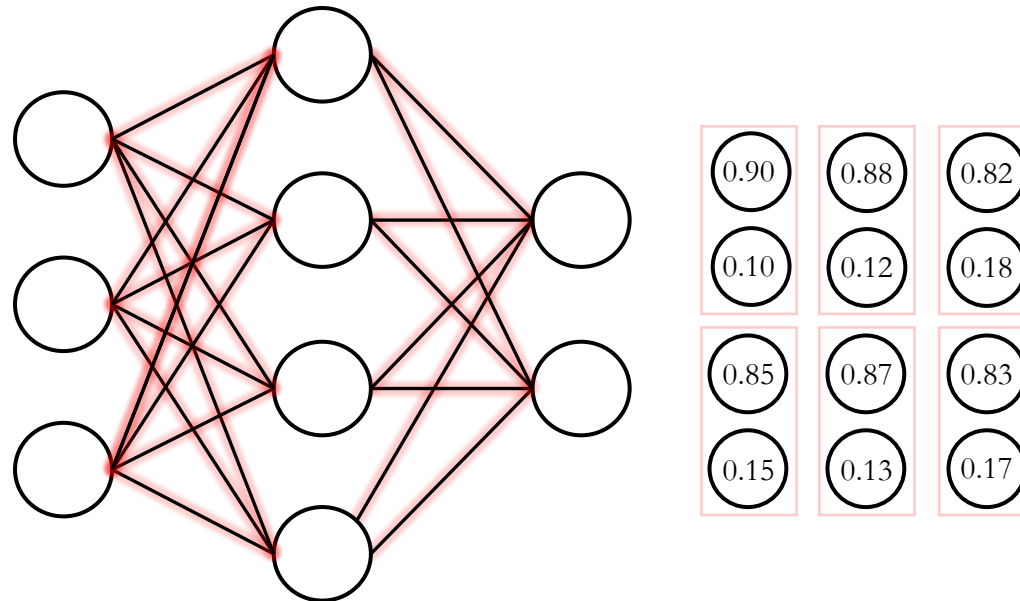
$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Reparameterization Trick: $w_{ij} = \mu_{ij} + \sigma_{ij}\epsilon ; \epsilon \sim \mathcal{N}(0, 1)$

$$\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)} = \begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix} + \begin{bmatrix} \sigma_{11}^{(1)} & \cdots & \sigma_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{(1)} & \cdots & \sigma_{HN}^{(1)} \end{bmatrix} \cdot \epsilon$$

Inference

- At inference time, you run the feedforward network for each sample N times, effectively thought of a N ensembles of neural networks
- At each feedforward step: $w_{ij}^l = \mu_{ij}^l + \sigma_{ij}^l \epsilon ; \epsilon \sim \mathcal{N}(0, 1)$



Other benefits...

- Variational continual learning (ICLR 2018) for sequential multiclass, multitask learning
 - Updating the prior based on the posterior of the previously learned task/class

$$\mathcal{L}_{ELBO} = -\mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_{\phi}(\theta) || p(\theta))$$

$$\mathcal{L}_{VCL}^t(q_t(\theta)) = -\mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_t(\theta) || q_{t-1}(\theta))$$

Other approaches...

